

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250283582

Kind Code

A1

Publication Date

September 11, 2025

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BULB-TYPE LIGHT SOURCE

Abstract

A light device including a cover, a light emitter including a first end, a second end, and an intermediate region, and a power board, in which the light emitter includes light sources including first and second light sources adjacent to the first end and the second end, respectively, 5 each of the light sources including first and second contact electrodes, first and second electrode pads electrically connected to at least one of the first and second contact electrodes, respectively, a wavelength converter, and a connection line disposed on an insulation layer and electrically connected to the first or second electrode pad that is electrically connected to the light sources, in which a first distance between the first light source and a first region of the cover is different 10 from a second distance between the second light source and a second region of the cover.

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Family ID: 1000008619517

Appl. No.: 19/215595

Filed: May 22, 2025

Foreign Application Priority Data

KR 1020170155075

Nov. 20, 2017

Related U.S. Application Data

parent US continuation 18739674 20240611 parent-grant-document US 12345381 child US 19215595

parent US continuation 18363081 20230801 parent-grant-document US 12038141 child US 18739674

parent US continuation 18067286 20221216 parent-grant-document US 11761591 child US

18363081

parent US continuation 17385327 20210726 parent-grant-document US 11530783 child US 18067286

parent US continuation 16822312 20200318 parent-grant-document US 11073249 child US 17385327

parent WO continuation PCT/KR2018/013157 20181101 PENDING child US 16822312

Publication Classification

Int. Cl.: F21K9/235 (20160101); F21K9/232 (20160101); F21Y115/10 (20160101)

U.S. Cl.:

CPC F21K9/235 (20160801); F21K9/232 (20160801); F21Y2115/10 (20160801)

Background/Summary

CROSS REFERENCE OF RELATED APPLICATIONS [0001] The application is a continuation of U.S. patent application Ser. No. 18/739,674, filed on Jun. 11, 2024, which is a continuation of U.S. patent application Ser. No. 18/363,081, filed on Aug. 1, 2023, now issued as U.S. Pat. No. 12,038,141, which is a continuation of U.S. patent application Ser. No. 18/067,286, filed on Dec. 16, 2022, now issued as U.S. Pat. No. 11,761,591, which is a continuation of U.S. patent application Ser. No. 17/385,327, filed on Jul. 26, 2021, now issued as U.S. Pat. No. 11,530,783, which is a continuation of U.S. Patent application Ser. No. 16/822,312, filed on Mar. 18, 2020, now issued as U.S. Pat. No. 11,073,249, which is a continuation of PCT Application No. PCT/KR2018/013157, filed on Nov. 1, 2018, which claims priority to and the benefits of Korean Patent Application No. 10-2017-0155075, filed on Nov. 20, 2017, each of which is hereby incorporated by reference for all purposes as if fully set forth herein.

BACKGROUND

Field

[0002] Embodiments of the invention relate generally to a bulb-type light source having a bulb shape, and more particularly, relate to a bulb-type light source including a light emitting device.

Discussion of the Background

[0003] Conventionally, a light bulb employing a filament as its light source is being used; however, the light bulb is being gradually replaced with a light emitting device, such as a light emitting diode, as the light bulb employing the filament has low light amount and high power consumption. When the light emitting device is used, the light emitting device is made to have the look and shape of a conventional filament for a decorative purpose. For instance, the light emitting device may be designed to have the same shape as a conventional filament bulb.

[0004] However, in the case where the light source having the same shape as the conventional filament bulb is manufactured using the light emitting device, a light emitting direction may be limited.

[0005] The above information disclosed in this Background section is only for understanding of the background of the inventive concepts, and, therefore, it may contain information that does not constitute prior art.

SUMMARY

[0006] A lighting device according to various embodiments of the present disclosure includes a substrate having a plurality of flat portions and a non-flat portion disposed between the flat

portions, a plurality of light emitting sources disposed on the substrate, a fluorescent substrate layer covering one or more light emitting sources and converting a wavelength of a light from the light emitting source, and a connection line disposed on the substrate and electrically connecting the light emitting sources adjacent to each other between the adjacent light emitting sources. The substrate has a first end and a second end are arranged at different distance from a central axis. [0007] Additional features of the inventive concepts will be set forth in the description which follows, and in part will be apparent from the description, or may be learned by practice of the inventive concepts.

[0008] In at least one variation, the lighting device further includes a globe to transmit a light emitted from the light emitting source.

[0009] In another variant, the lighting device further includes a socket engaged with the globe and a power board disposed in the socket and connected to the light emitting source.

[0010] In further another variant, the first end of the substrate is connected to the power board, and the second end of the substrate is spaced apart from the power board.

[0011] In another variant, the plurality of light emitting sources are arranged radially about a center of the globe when viewed from one direction.

[0012] In another variant, the first end of the substrate is farther away from the center than the second end of the substrate is.

[0013] In another variant, the substrate is farther away from the center at a position between the first and second ends than from the first and second ends.

[0014] In another variant, the lighting device further includes a fixing plate that fixes the second end.

[0015] In another variant, each light emitting source emits light with an orientation angle in two or more light emission areas, and two light emission areas are positioned at the second ends adjacent to each other and overlap with each other. A distance from the fixing plate to the overlapping area is smaller than a distance from the fixing plate to the globe.

[0016] In another variant, an electrode pad is disposed at the first end to supply power to the light emitting sources.

[0017] A lighting device according to embodiments of the present disclosure, includes a globe transmitting a light, a substrate comprising a plurality of flat portions and at least one of non-flat portion, a plurality of light emitting sources disposed on the substrate, a fluorescent substrate layer covering one or more light emitting sources and converting a wavelength of a light from the light emitting source, and a connection line disposed on the substrate and electrically connecting the light emitting sources adjacent to each other between the adjacent light emitting sources. The substrate has a first end and a second end, the first end and the second end arranged at different distance from a central axis.

[0018] In at least one variant, the lighting device further includes a socket engaged with the globe, and a power board disposed in the socket and connected to the light emitting source.

[0019] In another variant, the first end of the substrate is connected to the power board, and the second end of the substrate is spaced apart from the power board.

[0020] In further another variant, the plurality of light emitting sources are arranged radially about a center of the globe when viewed from one direction.

[0021] In another variant, the first end of the substrate is farther away from the center than the second end of the substrate is.

[0022] In another variant, the substrate is farther away from the center at a position between the first and second ends than from the first and second ends.

[0023] The lighting device further includes a fixing plate that fixes the second end.

[0024] In another variant, each light emitting source emits a light with an orientation angle in two or more light emission areas, and two light emission areas are positioned at the second ends adjacent to each other and overlap with each other. A distance from the fixing plate to the

overlapping area is smaller than a distance from the fixing plate to the globe.

[0025] In another variant, an electrode pad is disposed at the first end to supply a power to the light emitting sources.

[0026] In another variant, the power board is provided with an insertion hole into which the first end is inserted, and the power is supplied to the light emitting device chips through the electrode pad when the first end is inserted into the insertion hole.

[0027] Various embodiments disclosed in the present disclosure provide a bulb-type light source having the same shape as a conventional light bulb and having a light emitting device filament having good light uniformity in all directions.

[0028] A bulb-type light source according to various embodiments of the present disclosure includes a globe transmitting a light and at least one light emitting device filament disposed in the globe. The light emitting device filament includes a substrate including n (n is a natural number equal to or greater than 2) flat portions and $n-1$ bendable portions disposed between the flat portions, a plurality of light emitting device chips disposed on the flat portions, a fluorescent substance layer covering the light emitting device chip and converting a wavelength of a light from the light emitting device chip, and a connection line disposed on the flat portions and electrically connecting the light emitting device chips adjacent to each other between the adjacent light emitting device chips.

[0029] The substrate includes at least one notch defined in the bendable portions, and at least one light emitting device chip is disposed in each flat portion.

[0030] The connection line has different widths from each other at the bendable portion. The connection line has a zigzag shape at the bendable portion.

[0031] The fluorescent substance layer is disposed at, at least a portion of the flat portion and covers the light emitting device chips. The fluorescent substance layer extends from the flat portion and covers the bendable portion. The fluorescent substance layer is bent at the bendable portion. The fluorescent substance layer has different thicknesses in the flat portions and the bendable portion.

[0032] At least one flat portion among the flat portions has a length different from the other flat portions.

[0033] The notch is provided along a width direction of the light emitting device filament. The notch is provided along a longitudinal direction of the light emitting device filament. The notch is provided to be inclined with respect to a longitudinal direction of the light emitting device filament.

[0034] The bulb-type light source further includes a socket engaged with the globe and a power board disposed in the socket and connected to the light emitting device filament.

[0035] A first end of the light emitting device filament is connected to the power board, and a second end of the light emitting device filament is spaced apart from the power board.

[0036] The light emitting device filament is provided in plural, and the light emitting device filaments are arranged radially about a center of the globe when viewed from one direction.

[0037] The first end of the light emitting device filament is farther away from the center than the second end of the light emitting device filament is.

[0038] The light emitting device filament is farther away from the center at a position between the first and second ends than at the first and second ends.

[0039] The bulb-type light source further includes a fixing plate that fixes the second ends.

[0040] When each light emitting device chip emits a light with an orientation angle, light emission areas in the orientation angle of the light emitting device chips, which are positioned at the second ends adjacent to each other, overlap each other, and a distance from the fixing plate to the overlapping area is smaller than a distance from the fixing plate to the globe.

[0041] An electrode pad is disposed at the first end to supply a power to the light emitting device chips.

[0042] The power board is provided with an insertion hole into which the first end is inserted, and

the power is supplied to the light emitting device chips through the electrode pad when the first end is inserted into the insertion hole.

[0043] Other aspects, advantages, and salient features of the present disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the present disclosure.

[0044] According to an exemplary embodiment of the present disclosure, the bulb-type light source with the shape similar to that of the conventional bulb but with the high light uniformity in all directions is provided.

[0045] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are intended to provide further explanation of the invention as claimed.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0046] The accompanying drawings, which are included to provide a further understanding of the invention and are incorporated in and constitute a part of this specification, illustrate embodiments of the invention, and together with the description serve to explain the inventive concepts.

[0047] FIG. 1 is a perspective view showing a bulb-type light source according to an exemplary embodiment of the present disclosure.

[0048] FIG. 2A is a side cross-sectional view showing a bulb-type light source according to an exemplary embodiment of the present disclosure.

[0049] FIG. 2B is a top view showing a bulb-type light source according to an exemplary embodiment of the present disclosure.

[0050] FIG. 3A is a plan view showing a light emitting device filament according to an exemplary embodiment of the present disclosure.

[0051] FIG. 3B is a side view showing the light emitting device filament of FIG. 3A.

[0052] FIG. 3C is a side view showing the light emitting device filament of FIG. 3B having a bendable portion in a bent shape.

[0053] FIG. 4A is an enlarged side cross-sectional view showing a portion P1 of FIG. 3B.

[0054] FIG. 4B is an enlarged side cross-sectional view showing a portion P2 of FIG. 3B.

[0055] FIG. 4C is an enlarged side cross-sectional view showing a portion P2 of FIG. 3B with an additional line.

[0056] FIG. 5 is a cross-sectional view showing a light emitting device chip implemented using a light emitting diode according to an exemplary embodiment of the present disclosure.

[0057] FIG. 6A is a conceptual view showing a plurality of light emitting devices connected to a connection line in parallel according to an exemplary embodiment of the present disclosure.

[0058] FIG. 6B is a conceptual view showing a plurality of light emitting devices connected to a connection line in series according to an exemplary embodiment of the present disclosure.

[0059] FIG. 7 is a side cross-sectional view showing a portion of a bulb-type light source employing a light emitting device filament according to an exemplary embodiment of the present disclosure.

[0060] FIG. 8A is a perspective view showing a bendable portion and two flat portions adjacent to a bendable portion of a bulb-type light source according to an exemplary embodiment of the present disclosure.

[0061] FIG. 8B is a perspective view showing a bendable portion having a different shape from the bendable portion of FIG. 8A and two flat portions adjacent to the bendable portion of a bulb-type light source according to an exemplary embodiment of the present disclosure.

[0062] FIG. 9A is a cross-sectional view showing a first type of bendable portion.

[0063] FIG. 9B is a cross-sectional view showing a second type of bendable portion.
[0064] FIG. 9C is a cross-sectional view showing a third type of bendable portion.
[0065] FIG. 9D is a cross-sectional view showing a fourth type of bendable portion.
[0066] FIG. 9E is a cross-sectional view showing a fifth type of bendable portion.
[0067] FIG. 10A is a plan view showing a light emitting device filament according to another exemplary embodiment of the present disclosure.
[0068] FIG. 10B is a side view showing the light emitting device filament of FIG. 10A.
[0069] FIG. 10C is a side view showing the light emitting device filament of FIG. 10B in a bent shape.
[0070] FIG. 11A is a plan view showing a light emitting device filament according to another exemplary embodiment of the present disclosure.
[0071] FIG. 11B is a cross-sectional view taken along a line I-I' of FIG. 11A.
[0072] FIG. 12A is a plan view showing a light emitting device filament according to an exemplary embodiment of the present disclosure.
[0073] FIG. 12B is a view showing a bulb-type light source manufactured using the light emitting device filament of FIG. 12A.
[0074] FIG. 13A is a cross-sectional view showing a first type of globe.
[0075] FIG. 13B is a cross-sectional view showing a second type of globe.
[0076] FIG. 13C is a cross-sectional view showing a third type of globe.
[0077] FIG. 14 is a view showing a bulb-type light source according to an exemplary embodiment of the present disclosure.

DETAILED DESCRIPTION

[0078] In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of various embodiments or implementations of the invention. As used herein “embodiments” and “implementations” are interchangeable words that are non-limiting examples of devices or methods employing one or more of the inventive concepts disclosed herein. It is apparent, however, that various embodiments may be practiced without these specific details or with one or more equivalent arrangements. In other instances, well-known structures and devices are shown in block diagram form in order to avoid unnecessarily obscuring various embodiments. Further, various embodiments may be different, but do not have to be exclusive. For example, specific shapes, configurations, and characteristics of an embodiment may be used or implemented in another embodiment without departing from the inventive concepts.

[0079] Unless otherwise specified, the illustrated embodiments are to be understood as providing features of varying detail of some ways in which the inventive concepts may be implemented in practice. Therefore, unless otherwise specified, the features, components, modules, layers, films, panels, regions, and/or aspects, etc. (hereinafter individually or collectively referred to as “elements”), of the various embodiments may be otherwise combined, separated, interchanged, and/or rearranged without departing from the inventive concepts. The use of cross-hatching and/or shading in the accompanying drawings is

[0080] generally provided to clarify boundaries between adjacent elements. As such, neither the presence nor the absence of cross-hatching or shading conveys or indicates any preference or requirement for particular materials, material properties, dimensions, proportions, commonalities between illustrated elements, and/or any other characteristic, attribute, property, etc., of the elements, unless specified. Further, in the accompanying drawings, the size and relative sizes of elements may be exaggerated for clarity and/or descriptive purposes. When an embodiment may be implemented differently, a specific process order may be performed differently from the described order. For example, two consecutively described processes may be performed substantially at the same time or performed in an order opposite to the described order. Also, like reference numerals denote like elements.

[0081] When an element, such as a layer, is referred to as being “on,” “connected to,” or “coupled to” another element or layer, it may be directly on, connected to, or coupled to the other element or layer or intervening elements or layers may be present. When, however, an element or layer is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present. To this end, the term “connected” may refer to physical, electrical, and/or fluid connection, with or without intervening elements. Further, the D1-axis, the D2-axis, and the D3-axis are not limited to three axes of a rectangular coordinate system, such as the x, y, and z-axes, and may be interpreted in a broader sense. For example, the D1-axis, the D2-axis, and the D3-axis may be perpendicular to one another, or may represent different directions that are not perpendicular to one another. For the purposes of this disclosure, “at least one of X, Y, and Z” and “at least one selected from the group consisting of X, Y, and Z” may be construed as X only, Y only, Z only, or any combination of two or more of X, Y, and Z, such as, for instance, XYZ, XYY, YZ, and ZZ. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0082] Although the terms “first,” “second,” etc. may be used herein to describe various types of elements, these elements should not be limited by these terms. These terms are used to distinguish one element from another element. Thus, a first element discussed below could be termed a second element without departing from the teachings of the disclosure.

[0083] Spatially relative terms, such as “beneath,” “below,” “under,” “lower,” “above,” “upper,” “over,” “higher,” “side” (e.g., as in “sidewall”), and the like, may be used herein for descriptive purposes, and, thereby, to describe one elements relationship to another element(s) as illustrated in the drawings. Spatially relative terms are intended to encompass different orientations of an apparatus in use, operation, and/or manufacture in addition to the orientation depicted in the drawings. For example, if the apparatus in the drawings is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the exemplary term “below” can encompass both an orientation of above and below. Furthermore, the apparatus may be otherwise oriented (e.g., rotated 90 degrees or at other orientations), and, as such, the spatially relative descriptors used herein interpreted accordingly.

[0084] The terminology used herein is for the purpose of describing particular embodiments and is not intended to be limiting. As used herein, the singular forms, “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Moreover, the terms “comprises,” “comprising,” “includes,” and/or “including,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, components, and/or groups thereof, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. It is also noted that, as used herein, the terms “substantially,” “about,” and other similar terms, are used as terms of approximation and not as terms of degree, and, as such, are utilized to account for inherent deviations in measured, calculated, and/or provided values that would be recognized by one of ordinary skill in the art.

[0085] Various embodiments are described herein with reference to sectional and/or exploded illustrations that are schematic illustrations of idealized embodiments and/or intermediate structures. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments disclosed herein should not necessarily be construed as limited to the particular illustrated shapes of regions, but are to include deviations in shapes that result from, for instance, manufacturing. In this manner, regions illustrated in the drawings may be schematic in nature and the shapes of these regions may not reflect actual shapes of regions of a device and, as such, are not necessarily intended to be limiting.

[0086] As customary in the field, some embodiments are described and illustrated in the accompanying drawings in terms of functional blocks, units, and/or modules. Those skilled in the art will appreciate that these blocks, units, and/or modules are physically implemented by

electronic (or optical) circuits, such as logic circuits, discrete components, microprocessors, hard-wired circuits, memory elements, wiring connections, and the like, which may be formed using semiconductor-based fabrication techniques or other manufacturing technologies. In the case of the blocks, units, and/or modules being implemented by microprocessors or other similar hardware, they may be programmed and controlled using software (e.g., microcode) to perform various functions discussed herein and may optionally be driven by firmware and/or software. It is also contemplated that each block, unit, and/or module may be implemented by dedicated hardware, or as a combination of dedicated hardware to perform some functions and a processor (e.g., one or more programmed microprocessors and associated circuitry) to perform other functions. Also, each block, unit, and/or module of some embodiments may be physically separated into two or more interacting and discrete blocks, units, and/or modules without departing from the scope of the inventive concepts. Further, the blocks, units, and/or modules of some embodiments may be physically combined into more complex blocks, units, and/or modules without departing from the scope of the inventive concepts.

[0087] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure is a part. Terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and should not be interpreted in an idealized or overly formal sense, unless expressly so defined herein.

[0088] The present disclosure may be variously modified and realized in many different forms, and thus, specific embodiments will be exemplified in the drawings and described in detail hereinbelow. However, the present disclosure should not be limited to the specific disclosed forms, and be construed to include all modifications, equivalents, or replacements included in the spirit and scope of the present disclosure.

[0089] Hereinafter, embodiments of the present disclosure will be described in detail with reference to accompanying drawings. FIG. 1 is a perspective view showing a bulb-type light source according to an exemplary embodiment of the present disclosure. FIG. 2A is a side cross-sectional view showing a bulb-type light source according to an exemplary embodiment of the present disclosure. FIG. 2B is a top view showing a bulb-type light source according to an exemplary embodiment of the present disclosure.

[0090] Referring to FIGS. 1, 2A, and 2B, the bulb-type light source according to the exemplary embodiment of the present disclosure includes a globe **20** provided with an opening defined through one side thereof and formed of a light transmitting material, a light emitting device filament **10** disposed in the globe **20**, a power board **30** connected to one end of the light emitting device filament **10**, and a socket **40** connected to the opening of the globe **20** and the power board **30**.

[0091] The globe **20** has a spherical shape as a whole and has the opening for inserting the light emitting device filament **10**, which is defined through one side thereof. The globe **20** may have a complete or incomplete spherical shape except for a portion for the opening, but it should not be limited thereto or thereby. The globe **20** may have a variety of shapes, such as an oval shape, or a shape of which a portion protrudes. The opening may vary depending on the shape of the globe **20** and may have a circular or oval shape.

[0092] The globe **20** includes the light transmitting material to transmit a light emitted from the light emitting device filament **10**. Here, the phrase, “light transmitting” means that the globe **20** transmits at least a portion of the light emitted from the light emitting device filament **10** in the globe **20**, including the situations where the globe **20** is semi-transparent to transmit only a light having a specific wavelength, or only a portion of a light of a specific wavelength, or where the globe **20** is partially transparent. In addition, the phrase, “light transmitting” includes the situation where the globe **20** is transparent to transmit all the light. To this end, the globe **20** may be formed

of a transparent or semitransparent glass to transmit at least a portion of the light. However, a material for the globe **20** should not be limited thereto or thereby, and the globe **20** may be formed of a plastic material.

[0093] The light emitting device filament **10** is a filament-shaped component and includes a substrate and a light emitting device chip provided on the substrate. The light emitting device filament **10** emits the light. The light emitted from the light emitting device filament **10** transmits through the globe **20** that is transparent and then travels to the outside of the globe **20**.

[0094] The light emitting device filament **10** is provided in a form that is inserted into the globe **20** through the opening of the globe **20**. The light emitting device filament **10** may be provided in an elongated bar shape. One end in a longitudinal direction of the light emitting device filament **10** is referred to as a “first end” **111a** and the other end opposite to the first end **111a** in the longitudinal direction is referred to as a “second end” **111b**, as shown in FIG. 2A.

[0095] One or more light emitting device filaments **10** may be provided, for example, about two to about five. In the exemplary embodiment of the present disclosure, four light emitting device filaments **10** are provided. However, a number of the light emitting device filaments **10** should not be limited thereto or thereby, and the light emitting device filaments **10** may be provided in a greater number than the above-mentioned number depending on a size or brightness of the bulb-type light source.

[0096] Assuming that an imaginary line extending in a direction in which the socket **40** is engaged with the globe **20** is referred to as a “central axis” CL, the light emitting device filament **10** is engaged with the socket **40** and the power board **30** in the form of extending along the central axis CL. The light emitting device filament **10** may be bent multiple times and may have a bent shape such that a portion thereof is away from the central axis CL. That is, as shown in figures, the first end **111a** and the second end **111b** are arranged at a relatively small distance from the central axis CL. However, an area between the first end **111a** and the second end **111b** is arranged at a relatively large distance from the central axis CL. In addition, distances between the central axis CL and the light emitting device filament **10** at the first end **111a** and the second end **111b** may also be different from each other, and the distance between the central axis CL and the light emitting device filament **10** at the first end **111a** may be greater than the distance between the central axis CL and the light emitting device filament **10** at the second end **111b**.

[0097] A distance between the central axis CL and the light emitting device filament **10** closest to the central axis CL at the first end **111a** in a vertical direction of the central axis CL is referred to as a first distance d1, a distance between the central axis CL and the light emitting device filament **10** closest to the central axis CL at the second end **111b** in the vertical direction of the central axis CL is a second distance d2, and a distance between the central axis CL and the light emitting device filament **10** in the vertical direction of the central axis CL in the area between the first and second ends **111a** and **111b** is a third distance d3. The third distance d3 may be greater than the first distance d1 and/or the second distance d2. In addition, the first distance d1 may be greater than the second distance d2. In the exemplary embodiment, the light emitting device filament **10** is farthest away from the central axis CL in the area between the first end **111a** and the second end **111b**, so that an influence of heat between light emitting device filaments **10** adjacent to each other is minimized. In particular, as a diameter of the globe **20** is the largest in an area corresponding to the area between the first end **111a** and the second end **111b**, even though the heat is dissipated from the light emitting device filament **10**, not only the influence on the adjacent light emitting device filament **10** becomes small but also a heat dissipation toward the globe **20** disposed at an outer side of the light emitting device filament **10** is easy. Further, when the diameter of the globe **20** is larger than other areas, an amount of heat accumulated in a unit space area may be reduced. Accordingly, a heat dissipation effect is maximized by arranging the light emitting device filament **10** to be the farthest away from the central axis CL in the area between the first end **111a** and the second end **111b** where the diameter of the globe **20** is the largest.

[0098] In the exemplary embodiment of the present disclosure, as the third distance d3 is greater than the first distance d1 and/or the second distance d2, the transfer of heat adjacent to the light emitting device filament **10** may be minimized. In addition, since the first distance d1 is greater than the second distance d2, heat generated by the light emitting device filament **10** may be easily distributed or dissipated through the power board **30** (or a separate heat dissipation board provided under the power board **30**).

[0099] In the exemplary embodiment of the present disclosure, when the light emitting device filament **10** is provided in plural, the light emitting device filaments **10** may be arranged radially with respect to the central axis when viewed from the top perpendicular to the central axis CL. In addition, angles between the radially disposed light emitting device filaments **10** may be substantially equal to each other. Therefore, the light emitted from the light emitting device filament **10** may travel as uniformly as possible in all directions of 360 degrees with respect to the central axis CL. However, according to another exemplary embodiment of the present disclosure, in a case where an emission direction is required to be anisotropic, it is not necessary to arrange the light emitting device filaments **10** radially at equal intervals, and the light emitting device filaments **10** may be further arranged along a particular direction in which the light is directed.

[0100] The power board **30** is connected to the first end **111a** of the light emitting device filament **10**, and the power board **30** supplies a power to the light emitting device filament **10**. The second end **111b** of the light emitting device filament **10** is spaced apart from the power board **30**.

[0101] The power board **30** is disposed at the opening of the globe **20** and encapsulates the opening of the globe **20** together with the socket **40**. The power board **30** may have a variety of shapes and may fix the light emitting device filament **10** while supplying the power to the light emitting device filament **10**. In the exemplary embodiment of the present disclosure, the power board **30** may have a circular plate shape having a step difference. However, the shape of the power board **30** may vary depending on the shape and structure of the globe **20** and the socket **40** and the connection structure to the light emitting device filament **10**.

[0102] An insertion hole **31** may be defined in the power board **30**, and the first end **111a** of the light emitting device filament **10** may be inserted into and fixed to the insertion hole **31**. The insertion hole **31** may be provided in a number corresponding to the number of the light emitting device filaments **10**. A wire connection portion may be disposed inside the insertion hole **31** of the power board **30** to be electrically connected to an electrode pad disposed at the first end **111a** of the light emitting device filament **10**, and the wire connection portion is electrically connected to the electrode pad disposed at the first end **111a** of the light emitting device filament after the light emitting device filament **10** is inserted.

[0103] Although not shown in the figures such as FIGS. **1** and **2A-2B**, the power board **30** may further include a heat sink for the dissipation in a position adjacent to the electrode pad. The kind or shape of the heat sink should not be particularly limited and may have various known kinds or shapes.

[0104] A fixing plate **11** may be provided at the second end **111b** of the light emitting device filament **10**. The fixing plate **11** may be provided to allow the central axis CL to pass through a center portion thereof. When there are plural light emitting device filaments **10**, the fixing plate **11** may serve to collect and fix the second ends **111b** of the light emitting device filaments **10**. For example, the fixing plate **11** may have grooves in which the second ends **111b** are mounted, respectively, and the second ends **111b** may be stably supported with each other by being fitted in the grooves. Thus, although external impacts are applied to the bulb-type light source, a structure that is designed to be free of dark spots may be maintained as a gap between the second ends **111b** is not excessively widened or narrowed. When the light emitting device filaments **10** are not sufficiently stably fixed, one end of the elongated light emitting device filament **10** may be shaken when being applied with the external impact, and the light emitting device filament **10** may collide with another light emitting device filament adjacent thereto to apply a secondary shock, thereby

causing a defect, but the fixing plate **11** prevents the defect. However, when the light emitting device filament **10** is sufficiently firmly fixed and there is not much shaking due to the external impact, the fixing plate **11** may be omitted.

[0105] The socket **40** is engaged with the power board **30** and provided with an engaging member to be mounted on an external device (e.g., an electrical outlet). The socket **40** may include a screw thread contact **41** engaged with the opening of the glove **20** and an electrical foot contact **43** protruded downward. The screw thread contact **41** and the electrical foot contact **43** may include a conductive material, but it may be insulated from each other. In this case, the screw thread contact **41** and the electrical foot contact **43** are connected to the electrode pad of the light emitting device filament **10**, and the power is applied to the light emitting device filament **10** via the screw thread contact **41** and the electrical foot contact **43**. However, the screw thread contact **41** and the electrical foot contact **43** are not necessarily formed of the conductive material in the case where the screw thread contact **41** and the electrical foot contact **43** are made to resemble an old bulb merely for aesthetic appearance. In this case, a power supply formed of a conductive material, such as a connector, may be provided separately in addition to the screw thread contact and the electrical foot contact.

[0106] FIGS. **3A** to **3C** are views showing a light emitting device filament according to an exemplary embodiment of the present disclosure. FIG. **3A** is a plan view showing the light emitting device filament, and FIGS. **3B** and **3C** are side views showing the light emitting device filament.

[0107] FIG. **4A** is an enlarged side cross-sectional view showing a portion **P1** of FIG. **3B**, and FIGS. **4B** and **4C** are enlarged side cross-sectional views showing a portion **P2** of FIG. **3B**. For the convenience of explanation, FIGS. **3A** and **3B** show a state before the light emitting device filament is bent, and FIG. **3C** shows a state after the light emitting device filament is bent.

[0108] Referring to FIGS. **3A** to **3C** and **4A** to **4C**, the light emitting device filament according to the exemplary embodiment of the present disclosure is elongated in one direction and has the bar shape with a first end **111a** and a second end **111b**.

[0109] The light emitting device filament **10** includes the substrate **110** and the light emitting device chip **130** (FIGS. **4A** through **4C**) disposed on the substrate **110**. The substrate **110** has a bar shape elongated in one direction and is bent at least one time. The substrate **110** includes a conductive material, for example, a single metal of Al, Zn, Ag, W, Ti, Ni, Au, Mo, Pt, Pd, Cu, Cr or Fe, or an alloy thereof. In the exemplary embodiment of the present disclosure, the substrate **110** may include aluminum.

[0110] The substrate **110** includes a bendable portion **113** defined by bending a portion thereof and at least two flat portions **R1**, **R2**, **R3**, . . . , and **Rn** adjacent to each other with the bendable portion **113** disposed therebetween. Since the bendable portion **113** is disposed between the flat portions **R1**, **R2**, **R3**, . . . , and **Rn** adjacent to each other, the number of the flat portions **R1**, **R2**, **R3**, . . . , and **Rn** is one more than the number of the bendable portions **113**. In other words, when the number of the flat portions **R1**, **R2**, **R3**, **R4**, . . . , **Rn** is “**n**” (**n** is a natural number of two or more), the number of the bendable portions **113** is “**n-1**”.

[0111] A notch **115** having a shape recessed in a direction opposite to at least one surface of the substrate **110** is formed in the bendable portion **113**. In the present exemplary embodiment, the notch **115** may have a shape recessed along a width direction of the substrate **110** and may have a triangular shape in a cross-section.

[0112] Two or more bendable portions **113**, are provided in the substrate **110**. Although not shown in figures, when there are components, such as a film, formed on the bendable portion **113**, not only the substrate **110** of the bendable portion **113** but also other components on the bendable portion **113** have the bent shape.

[0113] In the exemplary embodiment of the present disclosure, the flat portions **R1**, **R2**, **R3**, . . . , **Rn** correspond to substantially a flat straight area. In the exemplary embodiment of the present disclosure, the flat portions **R1**, **R2**, **R3**, **R4**, . . . , and **Rn** may not be provided in a straight-line

shape, and at least some of the flat portions R1, R2, R3, R4, . . . , and Rn may be bent as needed. In this case, however, the substrate **110** may not be completely flexible and may include a relatively rigid material to have a flat shape.

[0114] As shown in FIG. 3B, the notch **115** is provided to allow the substrate **110** to be easily bent. In the present exemplary embodiment, the notch **115** may have a shape recessed along the width direction of the substrate **110**.

[0115] A thickness of the substrate **110** in the area where the notch **115** is formed is smaller than a thickness of the substrate **110** in the area where the notch **115** is not formed. Accordingly, the substrate **110** may be easily bent in the area where the notch **115** is formed.

[0116] The substrate **110** is configured to be more readily bendable toward the surfaces having the notch **115** formed therein according to the position and the shape of the notch **115**. This is because space formed from the notches **115** can be used to allow bending of the substrate **110**. Due to the notches **115**, the substrate **110** may not be configured to bend in the opposite direction, i.e., toward the surface that the notches **115** are not formed, as the substrate **110** is pulled away relative to the notches **115** for bending. In the exemplary embodiment of the present disclosure, the notch **115** may be defined in a lower surface of the substrate **110**, and the substrate **110** may be bent toward the lower surface thereof such that a portion of the lower surface becomes closer to another portion of the lower surface in a direction facing each other.

[0117] The flat portions R1, R2, R3, R4, . . . , and Rn may have a relatively flat shape compared with the bendable portion **113**. The number of the flat portions R1, R2, R3, R4, . . . , and Rn and the number of the notches **115** may be provided in one or more depending on the shape of the substrate **110** to be finally formed. In the present exemplary embodiment, four flat portions R1, R2, R3, R4, . . . , and Rn, i.e., n=4, and three bendable portions **113** are shown.

[0118] However, as described above, the number of the flat portions R1, R2, R3, R4, . . . , and Rn and the number of the bendable portions **113** should not be limited thereto or thereby and may be provided in various numbers. In the following drawings, a substrate including first, second, third, and fourth flat portions will be described as a representative example.

[0119] The flat portions R1, R2, R3, R4, . . . , and Rn may be sequentially arranged from the first end **111a**. The flat portions R1, R2, R3, R4, . . . , and Rn may have the same length as each other or may have different lengths from each other. That is, the length of at least one flat portion among the flat portions R1, R2, R3, R4, . . . , and Rn may have a value different from those of the other flat portions among the flat portions R1, R2, R3, R4, . . . , Rn. The length of each of the flat portions R1, R2, R3, R4, . . . , and Rn may vary depending on a final desired shape of the light emitting device filament **10**, i.e., how much bending of the light emitting device filament is desired. For example, in the exemplary embodiment of the present disclosure, when the lengths of the flat portions R1, R2, R3, R4, . . . , Rn are respectively referred to as lengths L1, L2, L3, L4, . . . , and Ln, the lengths L1, L2, L3, L4, . . . , and Ln may have values that sequentially decreases.

[0120] According to the lengths of the flat portions R1, R2, R3, R4, . . . , and Rn, the bent shape is changed, and an overall shape of the substrate **110** is changed. When the number of the bendable portions **113** provided in a predetermined area increases, the bending of the substrate **110** increases, and thus, a shape of the light emitting device filament **10** eventually becomes similar to a shape that has a small radius of curvature as a whole, though it is not a curved line. On the contrary, when the number of the bendable portions **113** provided in a predetermined area decreases, the shape of the substrate **110** becomes similar to a straight line and the light emitting device filament **10** has a large radius of curvature as a whole. Therefore, the shape of the light emitting device filament **10** may be changed as a whole by adjusting the number of the bendable portions **113** and the lengths L1, L2, L3, L4, . . . , and Ln of the flat portions R1, R2, R3, . . . , Rn.

[0121] In the exemplary embodiment of the present disclosure, the notches **115** between the flat portions R1, R2, R3, R4, . . . , and Rn are shown to have the same size, but they should not be limited thereto or thereby. According to another exemplary embodiment of the present disclosure,

each notch **115** may be provided in a different size according to an angle between two flat portions **R1**, **R2**, **R3**, **R4**, . . . , and **Rn** to be bent. For example, the recessed degree of one notch **115** from the lower surface of the substrate **110** and the width of the notch **115** may be smaller or larger than the other notches **115**.

[0122] According to the exemplary embodiment of the present disclosure, the shape and/or the bending angle of the light emitting device filament **10** may be controlled by adjusting the length of each of the flat portions **R1**, **R2**, **R3**, **R4**, . . . , and **Rn**, the number of the notches **115**, and the size of the notch **115**. The shape and/or the bending angle of the light emitting device filament **10** may be determined in consideration of the light emission direction and design factors, and the light emitting device filament **10** may be provided in a form corresponding to the shape of the globe **20**.

[0123] In the exemplary embodiment of the present disclosure, referring to a first flat portion **R1** closest to the first end **111a** and an n-th flat portion **Rn** (corresponding to a fourth flat portion **R4** in the drawing) closest to the second end **111b**, an angle formed between a longitudinal direction of the first flat portion **R1** and a longitudinal direction of the n-th flat portion **Rn** may be about 45 degrees to about 90 degrees.

[0124] The light emitting device chip **130** and a fluorescent substance layer **160** covering the light emitting device chip **130** are provided on at least a portion of the flat portions **R1**, **R2**, **R3**, **R4**, . . . , and **Rn**. In this case, the light emitting device chip **130** is not provided in the bendable portion **113**.

[0125] A connection line **150** is disposed on the flat portions **R1**, **R2**, **R3**, **R4**, . . . , and **Rn** and the bendable portion **113** to electrically connect the light emitting device chips **130** in the flat portions **R1**, **R2**, **R3**, **R4**, . . . , and **Rn**. The connection line **150** is for applying the power to the light emitting device chips **130**. The connection line **150** may be formed on the substrate **110** in various forms. The connection line **150** may be formed in a variety of different ways that allow the electrical connection, for example, in the form of a plating film, in the form of a wire, a solder paste, a connector, and the like.

[0126] The light emitting device chip **130** is disposed on at least some of the flat portions **R1**, **R2**, **R3**, **R4**, . . . , and **Rn** of the substrate **110** and emits a light. The light emitted from the light emitting device chip **130** may be an infrared light, an ultraviolet light, or a visible light, but it should not be particularly limited. In the exemplary embodiment of the present disclosure, the light emitting device chip **130** may emit the visible light.

[0127] The light emitting device chip **130** is disposed on at least two flat portions **R1**, **R2**, **R3**, **R4**, . . . , and **Rn** on the substrate **110**. One or more light emitting device chips **130** may be provided for each of the flat portions **R1**, **R2**, **R3**, **R4**, . . . , and **Rn** and may be provided in different numbers for each of the flat portions **R1**, **R2**, **R3**, **R4**, . . . , and **Rn**. The light emitting device chips **130** may be provided separately from each other and may be arranged in various forms on each flat portion of the substrate **110**. For example, the light emitting device chips **130** may be arranged in a line form, or a matrix form on the substrate **110** along a longitudinal direction of the substrate **110**. However, the arrangement of the light emitting device chips **130** should not be limited thereto or thereby and may be randomly arranged.

[0128] Each light emitting device chip **130** may emit lights having various colors. Each light emitting device chip **130** may include a variety of elements that emits the light, and a light emitting diode according to an exemplary embodiment of the present disclosure may be used.

[0129] Each light emitting device chip **130** may emit a white light and/or a color light. Each light emitting device chip **130** may emit one color; however, each light emitting device chip **130** may emit the white light and/or the color light by combining different colors. In the exemplary embodiment of the present disclosure, the light emitting device chip **130** may include a red light emitting device chip, a green light emitting device chip, and a blue light emitting device chip. However, the color emitted by the light emitting device chips **130** should not be limited thereto or thereby, and each light emitting device chip **130** may emit colors such as cyan, magenta, and yellow.

[0130] It is not necessary for each light emitting device chip **130** to use green, red, and/or blue light emitting device chips in order to implement the color, and a light emitting device chip **130** emitting a color other than the red, green, and blue light emitting device chips may be used.

[0131] The fluorescent substance layer **160** is disposed on the light emitting device chip **130**, covers the light emitting device chip **130**, and converts a wavelength of the light emitted from the light emitting device chip **130**.

[0132] For example, a red light emitting diode may be used to implement the red color; however, the red light may be emitted by using a blue or ultraviolet light emitting diode with the fluorescent substance layer **160** that absorbs a blue light or ultraviolet light and then emits the red color. In the same manner, a green light emitting diode may be used to implement the green color; however, the green light may be emitted by using the blue or ultraviolet light emitting diode with the fluorescent substance layer **160** that absorbs the blue light or ultraviolet light and then emits the green color.

[0133] The fluorescent substance layer **160** may be disposed on the substrate **110** and may cover at least one light emitting device chip **130**. In other words, the fluorescent substance layer **160** may be provided to cover each light emitting device chip **130**, or may be provided to cover two or more light emitting device chips **130**. In the exemplary embodiment of the present disclosure, the fluorescent substance layer **160** is provided to cover the light emitting device chips **130** disposed on each flat portions **R1**, **R2**, **R3**, . . . , and **Rn**; however, the present disclosure should not be limited thereto or thereby. Two fluorescent substance layers **160** spaced apart from each other may be disposed on each one of the flat portions **R1**, **R2**, **R3**, . . . , and **Rn**. In this case, however, the light emitting chips **130** may be covered by the fluorescent substance layer **160** from the top of the light emitting chips **130**. According to the exemplary embodiment of the present disclosure, if necessary, the fluorescent substance layer **160** may not be disposed on the light emitting device chip **130**.

When the fluorescent substance layer **160** is not provided on the light emitting device chip **130**, a separate structure for protecting the light emitting device chip **130** may be added; however, it should not be limited thereto or thereby.

[0134] The fluorescent substance layer **160** may not be greatly restricted in shape as long as the fluorescent substance layer **160** covers the light emitting device chip **130** and converts the color of the emitted light; however, the fluorescent substance layer **160** may be provided only on an upper surface of the substrate **110** in the exemplary embodiment of the present disclosure.

[0135] When the fluorescent substance layer **160** is provided on the upper surface of the substrate **110**, in particular, only on a portion of the upper surface of the substrate **110**, a lower surface and a side surface of the substrate **110** and/or a portion of the upper surface of the substrate **110** are exposed to the outside, and thus, the dissipation of heat through the substrate **110** may be easily performed. Accordingly, the heat generated by the light emitting device chip **130** may be effectively removed, and thus, a defective rate of the light emitting device chip **130** may be reduced.

[0136] In the above-described embodiments, for the convenience of explanation, only the light emitting device chip **130**, the connection line **150**, and the fluorescent substance layer **160** are shown on the substrate **110**, however, other additional components may be further provided. For example, an insulating adhesive layer may be further provided between the substrate **110** and the light emitting device chip **130** to attach the light emitting device chip **130** to the substrate **110**. The connection line **150** may be disposed on the insulating adhesive layer, and a photo solder resist (PSR) may be further disposed between the connection line **53** and the fluorescent substance layer **160**.

[0137] The light emitting device chip **130** according to the exemplary embodiment of the present disclosure may be implemented in various ways, and FIG. 5 is a cross-sectional view showing the light emitting device chip **130** implemented as a light emitting diode according to an exemplary embodiment of the present disclosure. The light emitting diode may be configured in various forms such as a vertical type or a flip type. In the present exemplary embodiment, the vertical-type light

emitting diode is shown as a representative example. However, the structure of the light emitting diode should not be limited thereto or thereby, and the following drawings should be understood as one embodiment of the present disclosure.

[0138] Referring to FIG. 5, the light emitting device chip **130** is provided on the substrate **110** with the insulating adhesive layer **120** interposed therebetween.

[0139] The light emitting device chip **130** may include a device substrate **131**, a first conductive type semiconductor layer **133**, an active layer **135**, a second conductive type semiconductor layer **139**, and first and second contact electrodes **140a** and **140b**.

[0140] The device substrate **131** is a growth substrate **110** to grow a III-V nitride-based semiconductor layer, and may be, for example, a sapphire substrate, particularly a patterned sapphire substrate. The device substrate **131** is preferably an insulating substrate; however, it should not be limited to the insulating substrate.

[0141] The first conductive type semiconductor layer **133**, the active layer **135**, and the second conductive type semiconductor layer **139** are sequentially disposed on the device substrate **131**. The first conductive type and the second conductive type have opposite polarities to each other. When the first conductivity type is an n-type, the second conductive type is a p-type, or alternatively, when the first conductive type is the p-type, the second conductive type is the n-type. In the exemplary embodiment of the present disclosure, a structure in which the n-type semiconductor layer, the active layer, and the p-type semiconductor layer are sequentially formed on the device substrate **131** will be described as a representative example.

[0142] The n-type semiconductor layer, the active layer, and the p-type semiconductor layer may be formed of a III-V nitride-based semiconductor, for example, a nitride-based semiconductor such as (Al, Ga, In)N. The n-type semiconductor layer, the active layer, and the p-type semiconductor layer may be formed by being grown on the substrate **110** in a chamber using a known method such as a metal-organic chemical vapor deposition (MOCVD). In addition, the n-type semiconductor layer includes n-type impurities (e.g., Si, Ge, or Sn), and the p-type semiconductor layer includes p-type impurities (e.g., Mg, Sr, or Ba). For example, the n-type semiconductor layer may include GaN or AlGaN containing Si as a dopant, and the p-type semiconductor layer may include GaN or AlGaN containing Mg as a dopant. Although the n-type semiconductor layer and the p-type semiconductor layer are each shown as having a single-layer structure in the drawings, these layers may have a multi-layer structure and may also include a superlattice layer. The active layer may have a single quantum well structure or a multi-quantum well structure, and a composition ratio of the nitride-based semiconductor is adjusted to emit a desired wavelength. For example, the active layer may emit a blue light, or an ultraviolet light.

[0143] The first contact electrode **140a** is disposed on the first conductive type semiconductor layer **133** on which the active layer **135** and the second conductive type semiconductor layer **139** are not provided, and the second contact electrode **140b** is disposed on the second conductive type semiconductor layer **139**.

[0144] The first and/or second contact electrodes **140a** and **140b** may have a single-layer, or multi-layer structure of metals. As the material of the first and/or second contact electrodes **140a** and **140b**, metals such as Al, Ti, Cr, Ni, Au, and alloys thereof may be used.

[0145] In the exemplary embodiment of the present disclosure, although the light emitting device chip **130** is briefly described with reference to the drawings, the light emitting device chip **130** may further include a layer having additional functions in addition to the above-described layer. For example, an electron blocking layer **137** may be disposed on the active layer **135**. The electron blocking layer is disposed between the active layer **135** and the second conductive type semiconductor layer **139** and prevents electrons that are not combined with holes in the active layer due to a relatively high energy band gap from being diffused to the second conductive type semiconductor layer **139** disposed thereon. The electron blocking layer **137** may include, for example, aluminum gallium nitride (AlGaN). Further, the light emitting device chip **130** may

include various layers, such as a reflective layer for reflecting a light, an additional insulating layer for insulating a specific component, and a solder-preventing layer for preventing the solder from being diffused.

[0146] When a forward bias is applied to the light emitting diode having the above-mentioned structure, the electrons are combined with the holes in the active layer **135**, and the light is emitted.

[0147] Referring to FIGS. **3A** to **3C** and **4A** to **4C** again, the first contact electrode and the second contact electrode are connected to the connection line **150** through wires. A first power and a second power are applied to the connection line **150**. When the first power and the second power are applied to the first contact electrode and the second contact electrode through the wires, the light emitting device chip **130** is driven to emit a light.

[0148] In the exemplary embodiment of the present disclosure, an electrode pad **170** connected to the connection line **150** is disposed at the first end **111a** (see FIGS. **3A** through **3C**). The electrode pad **170** includes a first electrode pad **170a** and a second electrode pad **170b**, which are respectively connected to the first contact electrode and the second contact electrode of the light emitting device chip **130**. The first and second electrode pads **170a** and **170b** are disposed at the first end **111a** of the substrate **110**. The first and second electrode pads **170a** and **170b** are disposed to be spaced apart from each other, and upper surfaces of the first and second electrode pads **170a** and **170b** are exposed to the outside.

[0149] In the exemplary embodiment of the present disclosure, the first and second electrode pads **170a** and **170b** disposed at the first end **111a** may be engaged with the power board **30** in various forms to be electrically connected to the power board **30**.

[0150] For example, as shown in FIG. **4B**, the upper surface of the electrode pad **170** is exposed to the outside, the exposed upper surface makes contact directly with the wire connection portion provided in the insertion hole of the power board **30** when being inserted in the insertion hole, and thus, the power may be applied to the electrode pads **170**.

[0151] In the exemplary embodiment of the present disclosure, the electrode pad **170** may be provided in other forms. For example, it may be directly attached to the exposed first and second electrode pads using a conductive adhesive member (e.g., an anisotropic adhesive, a solder, etc.).

[0152] As shown in FIG. **4C**, an additional line **173** may be attached directly to the second electrode pad **170**. Referring to FIG. **4C**, a conductive adhesive member **175** may be disposed on the second electrode pad **170**, and the additional line **173** may be connected to the electrode pad **170** with the conductive adhesive member **175** interposed therebetween. The additional line **173** may have various shapes, such as a wire form or a plate shape. According to the present exemplary embodiment, the electrical connection with other components may be facilitated through the additional line **173**.

[0153] In the exemplary embodiment of the present disclosure, the light emitting device chips may be connected to the connection line in various forms. FIGS. **6A** and **6B** are conceptual views showing the light emitting device chips connected to the connection line according to an exemplary embodiment of the present disclosure.

[0154] Referring to FIG. **6A**, a plurality of light emitting device chips **130** may be connected to a connection line **150** in parallel. According to the present exemplary embodiment, first contact electrodes of each light emitting device chip **130** may be connected to the connection line **150** using a wire or other connection members, and second contact electrodes of each light emitting device chip **130** may be connected to the connection line **150** using a wire or other connection members.

[0155] Referring to FIG. **6B**, a plurality of light emitting device chips **130** may be connected to a connection line **150** in series. According to the present exemplary embodiment, the light emitting device chips **130** and the connection line **150** may be connected in series by connecting the connection line **150** connected to an electrode pad **170** to a first contact electrode of an adjacent light emitting device chip **130** and by connecting a second contact electrode of the adjacent light

emitting device chip **130** to a first contact electrode of an adjacent light emitting device chip **130** via a connection line **150** on the opposite side.

[0156] Although not shown in figures, different from the embodiments shown in FIGS. **6A** and **6B**, the light emitting device chip **130** may be arranged in a combination of serial and parallel connections. The connection method of the light emitting device chips **130** may be modified in various ways by taking into account a current applied thereto or an amount of light emitted therefrom. In addition, the connection line **150** is entirely connected to the light emitting device chips **130**, but it should not be limited thereto or thereby. That is, a plurality of connection lines **150** to which a power is individually supplied may be separately provided. For example, in the exemplary embodiment of the present disclosure, all the light emitting device chips **130** provided in the first, second, third, and fourth flat portions **R1**, **R2**, **R3**, and **R4** are directly or indirectly connected to each other. However, according to another embodiment, only the light emitting device chips **130** corresponding to each one of the flat portions **R1**, **R2**, **R3**, **R4**, . . . , and **Rn**, i.e., the same flat portion, may be directly or indirectly connected to each other, and the light emitting device chips **130** corresponding to different flat portions **R1**, **R2**, **R3**, **R4**, . . . , and **Rn** may be individually operated without being connected to each other.

[0157] In the exemplary embodiment of the present disclosure, as the substrate of the filament is formed not of a glass or ceramic material but of a metal material, the heat generated by the light emitting device chip may be effectively dispersed. Accordingly, due to the heat dissipation effect of the light emitting device chip, deterioration of the light emitting device chip may be prevented. In a conventional bulb-type light source, when a substrate used for the light emitting device filament is formed of the glass or ceramic material, it is necessary to fill a helium gas as a heat transfer medium in the globe for the heat dissipation. However, according to the exemplary embodiment of the present disclosure, the heat dissipation effect may be increased by using the metal substrate, and thus, there is no need to fill a heat dissipation gas, such as the helium gas, in the globe. Therefore, a manufacturing method of the bulb-type light source is simplified, and a manufacturing cost of the bulb-type light source is reduced. In addition, when the substrate is formed of the metal material, the light emitted from the light emitting device chip may be effectively scattered and/or reflected, and thus, the light emission efficiency may be improved.

[0158] As described above, as the bulb-type light source according to the exemplary embodiment of the present disclosure includes the light emitting device filament that is bent multiple times, the light emission area is significantly expanded, compared with the conventional bulb-type light source.

[0159] Even when a light emitting device filament is used to emit the light in a conventional bulb-type light source, the light emitting device filament of the conventional bulb-type light source is provided in a straight-line shape. In the case of the light emitting device filament provided in the straight-line shape, the light is emitted only in one direction that is perpendicular to the straight line. As a result, the light is hardly emitted in a direction parallel to the straight line, that is, the light is hardly emitted to both ends in a direction in which the light emitting device filament extends. Consequently, in the light emitting device filament of the straight-line shape, there are areas in which the amount of the light significantly decreases along a predetermined direction.

[0160] However, since the light emitting device filament according to the exemplary embodiment of the present disclosure includes the bendable portion, the light emitting device filament may be bent in various directions, and the light may be emitted in various directions rather than one direction. In addition, in the exemplary embodiment of the present disclosure, as the light emitting device filament is bendable, curved, or flexed in various directions or a plurality of bent light emitting device filaments is provided, the light may be uniformly emitted in all directions without reducing the amount of light in a specific direction. In particular, when the light emitting device filament according to the exemplary embodiment of the present disclosure is used, the amount of the light traveling upward along the extension direction of the central axis may be significantly

increased.

[0161] FIG. 7 is a side cross-sectional view showing a portion of a bulb-type light source employing a light emitting device filament according to an exemplary embodiment of the present disclosure with a traveling direction of a light emitted from a light emitting device in the light emitting device filament.

[0162] Referring to FIG. 7, the bulb-type light source according to the exemplary embodiment includes the bendable portion, and thus, the light emitting device filament **10** may be bent in various directions. Particularly, since the second end **111b** of the bendable portion is bent through the notch, the extension direction thereof may be close to the direction perpendicular to the direction in which the central axis passes. Accordingly, the emission angle of the light emitted from light emitting devices in the light emitting device filament **10** is also changed in accordance with the arrangement direction of the second end **111b** of the bendable portion. For example, when the light emitting device disposed on the second end **111b** emits a light having a predetermined orientation angle θ , the light emitted from the second end **111b** of the light emitting device filament **10** travels to an area shown in FIG. 7, i.e., to an upper portion and an area slightly inclined from the upper portion (an area within an angle indicated by θ). Here, when the light emitting device filament **10** is provided in plural numbers, there exists an area in which traveling directions of the light emitted from the light emitting device filaments **10** overlap each other, and dark spots are not visible since a sufficient amount of light travels in the overlapping area. Particularly, according to the exemplary embodiment of the present disclosure, when lights emitted from adjacent light emitting device filaments **10** facing each other travel symmetrically with an orientation angle of θ , a distance **a1** from the fixing plate **11** to the overlapping area of the lights emitted from the two filaments is smaller than a distance **a2** from the fixing plate **11** to the globe **20**, and thus, the dark spots in the upward direction of the bulb-type light source practically rarely occur.

[0163] This is explained in more detail as follows. In general, since the light emitting device filament is mounted on the substrate in a state where an individual light emitting device chip mounted on the light emitting device filament has a predetermined orientation angle (for example, from about 90 degrees to less than about 180 degrees, or from about 120 to about 150 degrees), the emission direction of the light may vary depending on the shape of the light emitting device filament. In the case where the light emitting device filament has the straight-line shape as the conventional light emitting device filament, the light emitted from the light emitting device chips is emitted only in a certain direction. As a result, the amount of the light is maximum in the direction perpendicular to the extension direction of the light emitting device filament, and the amount of the light is significantly reduced in the extension direction of the light emitting device filament and the opposite direction to the extension direction of the light emitting device filament.

[0164] Although the light emitting device filament according to the exemplary embodiment of the present disclosure includes the light emitting device chip having the predetermined orientation angle, the traveling direction of the light from the individual light emitting device chip in the light emitting device filament is variously expanded since the light emitting device filament is bent at various angles.

[0165] In other words, the light emitting device filament of the present disclosure has the bendable or curved shape at multiple positions to be substantially perpendicular or to be inclined to the extension direction of the central axis when the light emitting device filament extends along the central axis. The light emitting device chips are arranged along the bendable shape and emit the light entirely from the extension direction of the central axis to the opposite direction to the extension direction with respect to the central axis. Accordingly, the light emitting device filament of the present disclosure may provide the substantially uniform light from front to back along the extension direction of the central axis and over substantially 360 degrees along the direction perpendicular to the extension direction of the central axis.

[0166] Consequently, as the light emitting device filaments are bendable or curved, the individual

light emitting device chips are directed in various directions and finally emit the light while covering a large area. In this case, the degree of bending of the substrate at the bendable portion may be adjusted according to the orientation angle of the individual light emitting device chip. [0167] In addition, since the light emitting device filament according to the exemplary embodiment of the present disclosure may be formed of a relatively rigid material to have the flat shape at the flat portion other than the bent portion, it is easy to control the curvature in the overall shape. When the substrate is formed of a flexible material, it is difficult to control the angle at which the light emitting device filament is bent, and thus, it is not easy to form the light emitting device filament into a desired shape. In addition, when the substrate is formed of the flexible material, the substrate is easily deformed due to external impacts. In order to easily control the curvature while using the rigid substrate, the thickness of the substrate may be reduced. However, when the thickness of the substrate is reduced by more than a predetermined degree, the amount of heat that can dissipate through the substrate is also reduced, thereby decreasing the heat dissipation effect.

[0168] However, as the light emitting device filament according to the exemplary embodiment of the present disclosure may be bent at an appropriate degree by the bendable portion, the overall shape may be controlled. In addition, since the light emitting device filament is not flexible, the shape of the light emitting device filament is also less likely to be deformed by the external impacts. Further, the substrate is easily bendable while having the thickness with a predetermined degree or more due to the bendable portion, and the shape of the light emitting device filament may be easily changed and the heat dissipation effect may be improved

[0169] According to the exemplary embodiment of the present disclosure, the bulb-type light source having the above-described shape may employ a structure for preventing disconnection at the bendable portion.

[0170] FIGS. **8A** and **8B** are perspective views showing a bendable portion **113** and two flat portions adjacent to the bendable portion **113** of a bulb-type light source according to an exemplary embodiment of the present disclosure. FIGS. **8A** and **8B** show that the substrate **110** is bent at the bendable portion **113**, and some components are omitted. Hereinafter, different features from the above-mentioned features will be mainly described in order to avoid redundancy.

[0171] Referring to FIGS. **8A** and **8B**, the substrate **110** includes the bendable portion **113** in which a notch **115** is formed and the flat portions disposed at both sides of the bendable portion **113**.

[0172] A light emitting device chip **130** (not shown), a connection line **150** connecting each light emitting device chip (not shown) to adjacent light emitting device chip, and a fluorescent substance layer **160** covering the light emitting device chip **130** are formed at each flat portion of the substrate **110**. In the present exemplary embodiment, the light emitting device chip **130** and the fluorescent substance layer **160** are disposed only on each flat portion and not disposed on the bendable portion **113**.

[0173] The notch **115** is formed in a lower surface of the substrate **110**, and the substrate **110** is bent downward with respect to the notch **115** of the bendable portion **113**. As the substrate **110** is bent downward with respect to the bendable portion **113**, an upper surface of the substrate **110** corresponding to the bendable portion **113** is bent, and a tensile stress is applied to components disposed on the upper surface of the substrate **110**. Accordingly, lines disposed on the bendable portion **113** may be disconnected by the tensile stress.

[0174] In the exemplary embodiment of the present disclosure, in order to reduce the disconnection of the connection line **150** due to the tensile stress of the bendable portion **113**, a width, or a shape of the connection line **150** may be set differently. For example, as shown in FIG. **8A**, when the width of the lines in the bendable portion **113** increases, even though a defect such as a crack occurs in some portions of the lines, the disconnection may be prevented by connecting the lines by remaining portions of the lines. As another example, as shown in FIG. **8B**, the connection line **150** may have a zigzag shape in the bendable portion **113**. When the connection line **150** is disposed to be inclined to a direction to which the tensile stress is applied in the zigzag shape, the applied stress

is reduced. Therefore, the disconnection of the connection line **150** in the bendable portion **113** may be prevented.

[0175] According to the exemplary embodiment of the present disclosure, the bendable portion **113** may be modified in various ways to facilitate the bending of the light emitting device filament **10**. FIGS. **9A** to **9E** are cross-sectional views showing various types of bendable portions **113**. For the convenience of explanation, FIGS. **9A** to **9E** show only a substrate **110** and a notch **115**, and others are omitted.

[0176] Referring to FIGS. **9A** to **9E**, the bendable portion **113** may have various cross-sectional shapes. For example, the bendable portion **113** may have a semi-circular shape as shown in FIG. **9A** or may have a rectangular shape as shown in FIG. **9B**.

[0177] In addition, one notch **115** is provided in the bendable portion, however, as shown in FIG. **9C**, two or more notches **115** may be continuously arranged in the bendable portion.

[0178] Further, the notch **115** may be disposed on an upper surface as well as a lower surface. That is, since the surface provided with the notch **115** may be folded concavely or convexly in any direction, the notch **115** may be formed in a direction that is easy to bend. For example, the notch **115** may be disposed on the upper surface as shown in FIG. **9D**, or the notch **115** may be formed on both the upper surface and the lower surface as shown in FIG. **9E**.

[0179] As described above, the notch **115** may be provided in various shapes, various numbers, and various positions depending on the shape of the desired light emitting device filament.

[0180] In the exemplary embodiment of the present disclosure, the fluorescent substance layer may be provided in various shapes. FIGS. **10A** to **10C** are views showing a light emitting device filament according to another exemplary embodiment of the present disclosure. FIG. **10A** is a plan view showing the light emitting device filament, and FIGS. **10B** and **10C** are side views showing the light emitting device filament.

[0181] Referring to FIGS. **10A** to **10C**, in a light emitting device chip **130** according to another exemplary embodiment of the present disclosure, a fluorescent substance layer **160** is disposed on a bendable portion **113** and flat portions respectively disposed at both sides of the bendable portion **113**. In other words, the fluorescent substance layer **160** extends from the flat portion and is disposed on the bend portion **113**.

[0182] As a substrate **110** is bent at the bendable portion **113**, the fluorescent substance layer **160** disposed on the substrate **110** also has a bent shape at the bendable portion **113**. In this case, an upper surface of the substrate **110** is bent convexly, and the fluorescent substance layer **160** is bent in a convex shape along the upper surface of the substrate **110**. The fluorescent substance layer **160** is bent and subjected to a tensile stress at both sides thereof. Accordingly, the fluorescent substance layer **160** has a shape that stretches to the both sides at the bendable portion **113** and is thinner than the fluorescent substance layer **160** of each flat portion. Therefore, the fluorescent substance layer **160** has different thicknesses at the flat portion and the bendable portion **113**. In addition, although not shown in figures, an insulating adhesive, a connection line **150**, and a photo solder paste may be further provided between the substrate **110** and the fluorescent substance layer **160**. The insulating adhesive, the connection line **150**, and the photo solder paste have the bent shape at the bendable portion **113** and have different thicknesses at the flat portion and the bendable portion **113**.

[0183] In the exemplary embodiment of the present disclosure, the fluorescent substance layer **160** may include a material having elasticity, and thus, cutting or excessive bending of the fluorescent substance layer **160** may be reduced when the substrate **110** is bent.

[0184] According to the exemplary embodiment of the present disclosure, the bendable portion may be formed along a width direction as well as a longitudinal direction of the light emitting device filament. FIG. **11A** is a plan view showing a light emitting device filament according to another exemplary embodiment of the present disclosure, and FIG. **11B** is a cross-sectional view taken along a line I-I' of FIG. **11A**.

[0185] Referring to FIGS. **11A** and **11B**, the bendable portion may be bent in a width direction of a light emitting device filament, or alternatively in a longitudinal direction of the light emitting device filament. That is, the bendable portion may include a bendable portion **113** perpendicular to an extension direction of a substrate **110** and a bendable portion **113'** parallel to the extension direction of the substrate **110**. Accordingly, the substrate **110** of the light emitting device filament may be bent along both the longitudinal direction and the width direction of the substrate **110**. As the bendable portion is provided in various forms, a degree of freedom in the shape of the bendable portion increases.

[0186] In the present exemplary embodiment, a light emitting device chip **130** and a fluorescent substance layer may not be formed in areas in which the bendable portions **113** and **113'** are formed. In this case, the fluorescent substance layer may be divided into a first fluorescent substance layer **160a** and a second fluorescent substance layer **160b** spaced apart from each other with the bendable portion **113** disposed therebetween.

[0187] When plural fluorescent substance layers are provided and the fluorescent substance layers are spaced apart from each other, the fluorescent substance layers may convert a specific light into lights having the same color as each other, or may convert the specific light into lights having different colors from each other.

[0188] FIG. **12A** is a plan view showing a light emitting device filament according to an exemplary embodiment of the present disclosure, and FIG. **12B** is a view showing a bulb-type light source manufactured using the light emitting device filament of FIG. **12A**.

[0189] Referring to FIGS. **12A** and **12B**, a bendable portion **113''** may be disposed to be inclined with respect to a longitudinal direction or a width direction of a light emitting device filament **10** rather than the longitudinal direction or the width direction of the light emitting device filament **10**. When the bendable portion **113''** is inclined with respect to the longitudinal direction or the width direction, the light emitting device filament is bent in an oblique direction. Accordingly, a spiral-shaped filament may be implemented as shown in FIG. **12B**. The spiral-shaped filament may emit the light in various directions as well as a certain direction, and thus, uniformity of the light may be enhanced.

[0190] According to an exemplary embodiment of the present disclosure, a shape of the globe may be changed in various forms. FIGS. **13A** to **13C** are cross-sectional views showing various types of globes **20**.

[0191] Referring to FIGS. **13A** to **13C**, the globe **20** may have a radius that increases or decreases along a central axis as a distance from a socket **40** increases. Alternatively, the radius may be maintained and then may increase or decrease. The shape of the globe **20** may vary according to various designs. The shape of the light emitting device filament **10** may vary depending on the shape of the globe **20**. When the radius from the central axis of the globe **20** is large, the degree of bending of the light emitting device filament **10** may also increase, and when the radius from the central axis of the globe **20** is small, the degree of bending of the light emitting device filament **10** may also decrease.

[0192] According to an exemplary embodiment of the present disclosure, the connection relation between the light emitting device filament and the power board may be provided differently from the above-described embodiment. FIG. **14** is a view showing a bulb-type light source according to an exemplary embodiment of the present disclosure, and instead of forming an electrode pad at a first end **111a** of a light emitting device filament, a wire electrode **171** is provided. In the exemplary embodiment of the present disclosure, the wire electrode **171** may be connected to a line unit in a socket **40**. As described above, the light emitting device filament may be supplied with power from the line unit in various ways, and a power supply method may be changed in various ways as necessary.

[0193] According to the exemplary embodiment of the present disclosure, as the filament having the straight-line shape is easily bent using the notch, the light emitting device filament having

various shapes may be manufactured. In the exemplary embodiment of the present disclosure, when several filaments having the straight-line shape are provided, the several filaments having the straight-line shape may be bent simultaneously by fixing the second end at the angle “ θ ” and then pushing the first ends in the direction toward the second end simultaneously. For example, after inserting several filaments, which are not yet bent and have the straight-line shape, into the globe, the filament having the straight-line shape may be bent into a specific shape by pushing one end of the filament having the straight-line shape inwardly. In this case, since the light emitting device filament is bent in the globe, it is possible to manufacture the light emitting device filament that is wider than the opening of the globe.

[0194] According to the exemplary embodiment of the present disclosure, the bulb-type light source with the shape similar to that of the conventional bulb but with the high light uniformity in all directions is provided.

[0195] Although certain embodiments and implementations have been described herein, other embodiments and modifications will be apparent from this description. Accordingly, the inventive concepts are not limited to such embodiments, but rather to the broader scope of the appended claims and various obvious modifications and equivalent arrangements as would be apparent to a person of ordinary skill in the art.

Claims

1. A light device, comprising: a cover configured to transmit light; a light emitter disposed in the cover and including a first end, a second end, and an intermediate region disposed between the first end and the second end; and a power board configured to supply power to the light emitter, wherein the light emitter comprises: light sources including a first light source and a second light source adjacent to the first end and the second end, respectively, each of the light sources including a first contact electrode and a second contact electrode; a first electrode pad electrically connected to at least one of the first contact electrodes of the light sources; a second electrode pad electrically connected to at least one of the second contact electrodes of the light sources; a wavelength converter configured to convert a wavelength of light emitted from at least one of the light sources, and a connection line disposed on an insulation layer and electrically connected to the first electrode pad or the second electrode pad that is electrically connected to the light sources, wherein a first distance between the first light source and a first region of the cover is different from a second distance between the second light source and a second region of the cover.
